

## Research Interests

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**Autonomous driving, Robotics, Learning-based perception, Sensor fusion, Deep learning**

## Education

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**M.S. / Ph.D. in Electrical and Computer Engineering**

Mar 2017 - Feb 2023

Seoul National University

Vehicle Intelligence Laboratory, Supervisor: Prof. Seung-Woo Seo

**B.S. in Electrical and Computer Engineering**

Mar 2012 - Feb 2017

Seoul National University

## Publication

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**CIPER: A Unified Framework for Cross-view Image-retrieval and Pose-estimation**

Jeon, Yurim, Dongseong Seo, and Seung-Woo Seo, [Supplementary video](#)

arXiv Preprint, 2026

**EVT: Efficient View Transformation for Multi-Modal 3D Object Detection**

Lee, Yongjin, Hyeon-Mun Jeong, Yurim Jeon, and Sanghyun Kim

International Conference on Computer Vision (ICCV), 2025 (Acceptance Rate: 24%)

**Follow the Footprints: Self-supervised Traversability Estimation for Off-road Vehicle Navigation based on Geometric and Visual Cues**

Jeon, Yurim, E In Son, and Seung-Woo Seo, [Supplementary video](#)

IEEE International Conference on Robotics and Automation (ICRA), 2024 (Acceptance Rate: 45%)

**EFGHNet: A Versatile Image-to-Point Cloud Registration Network for Extreme Outdoor Environment**

Jeon, Yurim and Seung-Woo Seo, [Supplementary video](#)

IEEE Robotics and Automation Letters, vol. 7, no. 3 (2022) (Impact Factor: 5.2)

Presented in IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2022

**ABCD: Attentive Bilateral Convolutional Network for Robust Depth Completion**

Jeon, Yurim, Hwichang Kim, and Seung-Woo Seo, [Supplementary video](#)

IEEE Robotics and Automation Letters, vol. 7, no. 1 (2021) (Impact Factor: 4.3)

**A BRIEF-Gist based Efficient Place Recognition for Indoor Home Service Robots**

Jeon, Yurim, Taejae Lee, Chulhong Kim, Donghoon Yi, and Dong-Il Dan Cho

2016 16th International Conference on Control, Automation and Systems (ICCAS), 2016

## Work Experience

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**Staff Engineer**

May 2026 - Present

Samsung Electronics Future Robotics Office

- Developing a perception system for a humanoid robot.

**AI/ML Software Engineer**

Oct 2024 - Apr 2026

Boeing Korea

- Developed AI engines for automatic vision inspection.

**Research Scientist**

Dec 2023 - Sep 2024

Thordrive

- Developed a multi-sensor object detection module for urban autonomous driving.
- Developed an automatic label generation system for a large-scale object detection dataset.

## Postdoctoral Researcher

Mar 2023 - Nov 2023

Seoul National University

- Led a team in developing a perception system for off-road autonomous driving.
- Developed a traversable space estimation module for an unmanned ground vehicle deployed in off-road environments.

## Invited Talks

### Where the Sky Meets the Ground: UAV-UGV Collaborative Intelligence

Ajou University, Suwon, Korea

Dec 11, 2025

### Robot, Perception in Off-road

Ajou University, Suwon, Korea

Sep 30, 2024

## Research Project

### Development of perception system for unmanned vehicles in off-road scenarios

Government Project, Seoul National University

Jan 2022 - Nov 2023

- **Team leader**
- Designed hardware and configured sensors for unmanned ground vehicles.
- Developed a perception system for the off-road missions of unmanned robots.
- Optimized a perception system to operate on a low-end onboard computer of the robot.

### Research on human-level driving intelligence for autonomous driving of unmanned vehicles

Government Project, Seoul National University

Mar 2020 - Feb 2021

- Designed a selective perception system for autonomous driving, inspired by human recognition.
- Developed a neural layer using an attention mechanism as part of the selective perception system.
- Developed a neural network to generate high-resolution depth data from low-cost laser sensor data.

### Development of real-time object detection module using LiDAR sensor

Industry-Academia Cooperation Project, Seoul National University

Jan 2020 - Jun 2020

- **Project leader**
- Developed a deep learning object detection module using a LiDAR sensor for urban autonomous driving.
- Achieved strict criteria for accuracy and runtime in the context of urban driving scenarios.

## Technical Skills and Interests

### Language

**English** (Professional proficiency), **Korean** (Native proficiency)

### Developer Tools

Python (PyTorch, TensorFlow, etc.), C/C++ (ROS), Linux (Ubuntu), Git